CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

OX40 / 4-1BB / GITR: 5 molecules stopped early

5 stopped trials across 5 molecules: Adg-106, Gsk-3174998, Lipid Nanoparticle Encapsulated OX40L mRNA-2416, Tecaginlimab and 1 more. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

14.7%
5 of 34 closed trials (95% CI 6.4%–30.1%)

5 / 0 / 0
efficacy / safety / both (adds to 5)

39.5%
of 86 trials have closed · 52 still open or unresolved

4.9%
stopped for a biological reason within 3 years of starting (95% CI 1.6%–14.4%; 47 still at risk; 1.5% area-wide)

— **Completed and missed the primary endpoint.** None of the 15 completed trials in this segment posted a primary result on ClinicalTrials.gov that can be read, so nothing can be said here either way. 1 of the 12 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares

OX40 / 4-1BB / GITR
14.7%
5/34 closed

all oncology trials with a resolved drug target
5.2%
299/5739 closed

All oncology Phase 2,3
4.5%
353/7894 closed

What is in this class: 124 distinct experimental drugs across 86 trials (antibody 51, small molecule 27, unknown 21, bispecific antibody 13). Most tested: NIVOLUMAB (9), PEMBROLIZUMAB (9), UTOMILUMAB (8), AVELUMAB (7), CYCLOPHOSPHAMIDE ANHYDROUS (7). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 5 stops came from 5 sponsor-asset programmes across 5 sponsors; the largest (Pfizer) contributed 1; removing that programme's trials from both sides leaves 12.1%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 2 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 20.6%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	2	13	15.4%	4.3%–42.2%
2018–2020	1	12	8.3%	1.5%–35.4%
2021–2024	2	9	22.2%	6.3%–54.7%

Where the stops sit

Efficacy / futility: 5.
Sponsors with most stops: Genmab (1), Adagene (Suzhou) Limited (1), GSK (1), ModernaTX, Inc. (1), Pfizer (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05491317	2	Genmab GMAB	Tecaginlimab, PEMBROLIZUMAB	2025-08	Efficacy	After reviewing all available data from Part 1 of the trial, the Sponsor has decided not to proceed to Part 2 (randomized Phase 2) of the trial due to lack of efficacy
NCT04775680	2	Adagene (Suzhou) Limited ADAG	ADG-106, PD-1 antibody injection	2023-02	Efficacy	The safety of ADG106 combined with triprilimab has been fully understood.The overall clinical benefit of enrolled subjects was limited and further development of ADG106 was reconsidered.

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT06160609	2	GSK GSK	BELANTAMAB MAFODOTIN, GSK-3174998	2023-01	Efficacy	The study was terminated due to lack of efficacy.
NCT03323398	2	ModernaTX, Inc.	Lipid Nanoparticle Encapsulated O…, DURVALUMAB	2021-08	Efficacy	This study was halted prematurely because the efficacy endpoints were not met for either treatment arm.
NCT02951156	3	Pfizer PFE	AVELUMAB, UTOMILUMAB	2019-12	Efficacy	Study was terminated due to closure of study arms following futility analysis and difficulty in enrolling participants due to evolving treatment landscape

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 39.5% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.